

DATA DICTIONARY

File Name:	street_segment_crash_attributes.csv
Unit of Analysis:	Road segment (seg_id)
Time Period:	2020-2024 (5 years)
Total Segments:	41,259
Data Source:	PennDOT Crash Database

Data Processing:

Crashes were linked to street segments using a two-tier approach: (1) direct street name matching where possible, and (2) spatial snapping within 100 feet for crashes without matching street names.

Segment Identifiers

Variables that identify and describe road segments.

Variable	Type	Description	Range/Notes
seg_id	Integer	Unique segment identifier from Street Centerline shapefile	—
st_name	Character	Street name (e.g., MARKET)	—
st_type	Character	Street type suffix	ST, AVE, BLVD, etc.
full_street_name	Character	Complete street name (st_name + st_type)	—
class	Integer	Road functional classification	1-9
responsibl	Character	Responsible maintenance agency	STATE, CITY

Crash Counts

1. Total Counts

Variable	Type	Description	Range/Notes
total_crashes	Integer	Total crashes on segment (2020-2024)	0-200+
crashes_2020	Integer	Crashes in 2020	0-50+
crashes_2021	Integer	Crashes in 2021	0-50+
crashes_2022	Integer	Crashes in 2022	0-50+
crashes_2023	Integer	Crashes in 2023	0-50+
crashes_2024	Integer	Crashes in 2024	0-50+

2. Severity Counts

Variable	Type	Description	Range/Notes
ksi_crashes	Integer	Killed or Suspected Serious Injury crashes	0-30+
fatal_crashes	Integer	Fatal crashes (MAX_SEVERITY_LEVEL = 1)	0-5+
serious_inj	Integer	Suspected serious injury (MAX_SEVERITY_LEVEL = 2)	0-25+
minor_inj	Integer	Suspected minor injury (MAX_SEVERITY_LEVEL = 3)	0-50+
poss_inj	Integer	Possible injury (MAX_SEVERITY_LEVEL = 4)	0-50+
pdo_crashes	Integer	Property damage only (MAX_SEVERITY_LEVEL = 0)	0-100+

3. Vulnerable Road Users

Variable	Type	Description	Range/Notes
ped_crashes	Integer	Pedestrian-involved crashes	0-30+
bike_crashes	Integer	Bicycle-involved crashes	0-20+
vru_crashes	Integer	Vulnerable road user crashes (pedestrian + bicycle)	0-40+

4. Crash Characteristics

Variable	Type	Description	Range/Notes
speeding	<i>Integer</i>	Speeding-related crashes (flag set in crash database)	0-50+
alcohol	<i>Integer</i>	Alcohol-related crashes	0-20+
dark	<i>Integer</i>	Crashes in dark or unlit conditions	0-80+

Temporal Crash Distribution

1. Peak vs Off-Peak

Peak Hours Definition: 7-10 AM (AM peak) and 4-7 PM (PM peak)

Variable	Type	Description	Range/Notes
peak_crash	<i>Integer</i>	Crashes during peak hours (7-10 AM, 4-7 PM)	0-50+
offpk_crash	<i>Integer</i>	Crashes during off-peak hours	0-100+
am_peak	<i>Integer</i>	Crashes during AM peak (7-10 AM)	0-30+
pm_peak	<i>Integer</i>	Crashes during PM peak (4-7 PM)	0-30+
midday	<i>Integer</i>	Crashes during midday (10 AM-4 PM)	0-50+
evening	<i>Integer</i>	Crashes during evening (7-10 PM)	0-40+
night	<i>Integer</i>	Crashes during night/late hours (10 PM-7 AM)	0-50+

2. Day of Week

Variable	Type	Description	Range/Notes
weekend	<i>Integer</i>	Weekend crashes (Saturday-Sunday)	0-60+
weekday	<i>Integer</i>	Weekday crashes (Monday-Friday)	0-150+

Road Design Characteristics

These characteristics are extracted from crash records, not from a road inventory database. They represent the most commonly reported values in crashes on each segment.

Variable	Type	Description	Range/Notes
spd_lim_md	<i>Integer</i>	Mode (most common) speed limit reported in crashes on this segment, in mph	15-55
lane_ct_md	<i>Numeric</i>	Median lane count reported in crashes on this segment	1-8

Crash Rates and Proportions

All rates calculated as: $(count / total_crashes) \times 100$

1. Severity Rates

Variable	Type	Description	Range/Notes
ksi_rate	<i>Numeric</i>	Percentage of crashes that are KSI	0-100%
fatal_rate	<i>Numeric</i>	Percentage of crashes that are fatal	0-100%
ped_rate	<i>Numeric</i>	Percentage of crashes involving pedestrians	0-100%
bike_rate	<i>Numeric</i>	Percentage of crashes involving bicycles	0-100%
vru_rate	<i>Numeric</i>	Percentage of crashes involving vulnerable road users	0-100%

2. Crash Characteristic Rates

Variable	Type	Description	Range/Notes
spd_rate	<i>Numeric</i>	Percentage of crashes flagged as speeding-related	0-100%
alc_rate	<i>Numeric</i>	Percentage of crashes involving alcohol	0-100%
dark_rate	<i>Numeric</i>	Percentage of crashes in dark conditions	0-100%

3. Temporal Shares

Variable	Type	Description	Range/Notes
offpk_share	<i>Numeric</i>	Percentage of crashes (with time data) during off-peak hours. Formula: $(\text{offpk_crash} / (\text{peak_crash} + \text{offpk_crash})) \times 100$	0-100%
wknd_share	<i>Numeric</i>	Percentage of crashes on weekends. Formula: $(\text{weekend} / \text{total_crashes}) \times 100$	0-100%

Annual Crash Density

Annual crash rates calculated by dividing 5-year totals by 5.

Variable	Type	Description	Range/Notes
crash_py	<i>Numeric</i>	Average crashes per year $(\text{total_crashes} / 5)$	0-40+
ksi_py	<i>Numeric</i>	Average KSI crashes per year $(\text{ksi_crashes} / 5)$	0-6+